

Question 3: Scope Contour Diagrams

(a) Function + Loop Scope

main

└─ x = 2

└─ call f(x)

└─ y = 2

└─ x = y + 1 = 3

└─ for i in 0..2

└─ i

└─ temp = i*2 + x

(b) Reference Shadowing

main

└─ a = 5

└─ ref_To_An_A → a

└─ inner block

└─ a = 10 (shadows outer a)

└─ ref_To_An_A still points to outer a

(c) Pointer Scope

main

└─ p → heap(42)

└─ inner block

| └─ q = p
| └─ *q = 42
└─ delete p

(d) Same as (c), emphasized

main

|─ p → heap(42)
|─ inner block
| └─ q = p
| └─ *q = 42
└─ delete p